

JEFFREY WONG

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Education

University of Buffalo

Bachelor's of Science in Computer Science

Graduated: December 2025

Buffalo, NY

Relevant Coursework

- Computer Organization & Architecture
- Systems Engineering
- Data-Intensive Computing
- Programming Languages
- Machine Learning
- Distributed Systems
- Linear Algebra (Quantum Computing)
- Operating Systems (Concepts)

Technical Skills

Languages: C++, C, Python, Verilog, VHDL, Java, Javascript, CUDA

Frameworks/Technologies: GCC, Vim, Cmake, Git, Conan, Artifactory, Bash, PingDaVinci, InfluxDb, Docker

Experiences

CVS Health

May 2025 - August 2025

Software Engineer Intern (IAM / Security Engineering)

Hartford, CT

- Built distributed backend systems for identity orchestration within Ping Identity's ARIA platform, supporting high-volume authentication workflows across multi-tenant environments
- Designed and implemented RESTful API pipelines for SAML-based onboarding and identity lifecycle management, enabling scalable integration across federated systems
- Developed thread-safe OAuth 2.0 and JWT validation mechanisms, improving reliability and performance under high-concurrency authentication requests
- Engineered AI-driven automation pipelines using retrieval-augmented generation (RAG) to streamline identity configuration and reduce manual onboarding overhead
- Collaborated on secure authentication infrastructure, including token lifecycle management, access control policies, and adaptive authorization systems
- **Skills:** C++, Python, Distributed Systems, Concurrency, OAuth2, JWT, REST APIs, Authentication Systems

TrainAI

May 2024 - December 2025

Software Engineer

Buffalo, NY

- Built an AI-driven system for generating personalized nutrition plans using LLMs and structured biomedical datasets, processing user constraints and preferences at scale
- Designed backend services in Python (Flask) and Node.js to handle real-time request processing, data validation, and response generation
- Implemented data pipelines using Pandas and NumPy to compute macronutrient distributions and optimize dietary recommendations under user-defined constraints
- Developed interactive visualization tools for parameter tuning and real-time analysis of nutrition outputs
- **Skills:** Python, Data Processing, Backend Systems, APIs, NumPy, Pandas, MongoDB

Projects

Pipelined MIPS CPU Architecture & Cycle-Accurate Simulator

Fall 2025 - December 2025

- Designed a 5-stage pipelined MIPS processor (IF/ID/EX/MEM/WB) with hazard detection, data forwarding, and control logic to improve instruction throughput and pipeline efficiency
- Developed a cycle-accurate CPU simulator in C++, modeling pipeline stages, register file behavior, ALU execution, and control unit logic at the hardware-software boundary
- Simulated memory hierarchy behavior including stack/heap growth, address translation, and load/store operations, ensuring correctness against ISA-level specifications
- Analyzed pipeline performance and instruction-level parallelism (ILP), identifying bottlenecks related to hazards, stalls, and control flow
- **Skills:** C++, Computer Architecture, Pipeline Design, Instruction-Level Parallelism, VHDL, Quartus, ModelSim

Dynamic Memory Allocator

Fall 2024 - December 2024

- Designed and implemented a dynamic memory allocator (malloc, calloc, realloc, free) from scratch, managing heap blocks via metadata headers and boundary tags.
- Implemented best-fit and first-fit allocation strategies with coalescing to reduce external and internal fragmentation.
- **Skills:** C, C++, Linux, Systems Programming

Algorithm

September 2024 - December 2024

- Developed step-by-step visualizations for key algorithms such as BFS, DFS, Dijkstra's, and Binary Search, allowing users to explore and understand their operations in real-time.
- **Skills:** React.js, Node.js

Book Recommendation System

January 2024 - May 2024

- Used matplotlib to visualize data cleaned using EDA processes such as dropping rows / generating missing data with pandas library in Python to for users to enjoy niche personalized content for informed purchases.
- Utilized Machine learning algorithms such as Linear Regression for price predictions, K-Means Clustering for audiobook categorization, and Logistic Regression for generating recommendations
- **Skills:** Python, Pandas, Numpy, Matplotlib